Python Interview Guide

Fundamentals, Data Structures, OOP, Tricky Questions

1. Core Concepts

- Python is dynamically typed and interpreted (bytecode via CPython).
- GIL (Global Interpreter Lock) limits true multi-threading for CPU work.
- Lists vs Tuples: mutable vs immutable.
- Sets: unordered, unique. Dicts: key-value, insertion-ordered (3.7+).

2. Common Built-ins

- len(), range(), enumerate(), zip(), map(), filter(), sorted().
- Lambda: anonymous single-expression function.
- List comprehension: [x*2 for x in range(10) if x % 2 == 0].

3. OOP in Python

- Class vs Instance vs Static methods (@classmethod, @staticmethod).
- __init__ constructor, __str__ / __repr__ for display.
- Inheritance, Method Resolution Order (MRO), super().
- Multiple inheritance supported (unlike Java).

4. Decorators & Generators

- Decorator: function that wraps another function.
- Generator: yields values lazily using yield; memory efficient.

5. Error Handling

- try / except / else / finally blocks.
- Custom exceptions subclass Exception.

6. Memory & Performance

- Shallow vs deep copy (copy.copy vs copy.deepcopy).
- == compares values, is compares identity.
- String immutability: joining with join() is faster than +=.

7. Tricky Interview Questions

- Output of [1,2,3]*2 -> [1,2,3,1,2,3].
- Default mutable argument pitfall (use None instead of []).
- Difference between remove, pop, and del on lists.
- Global vs nonlocal keyword.